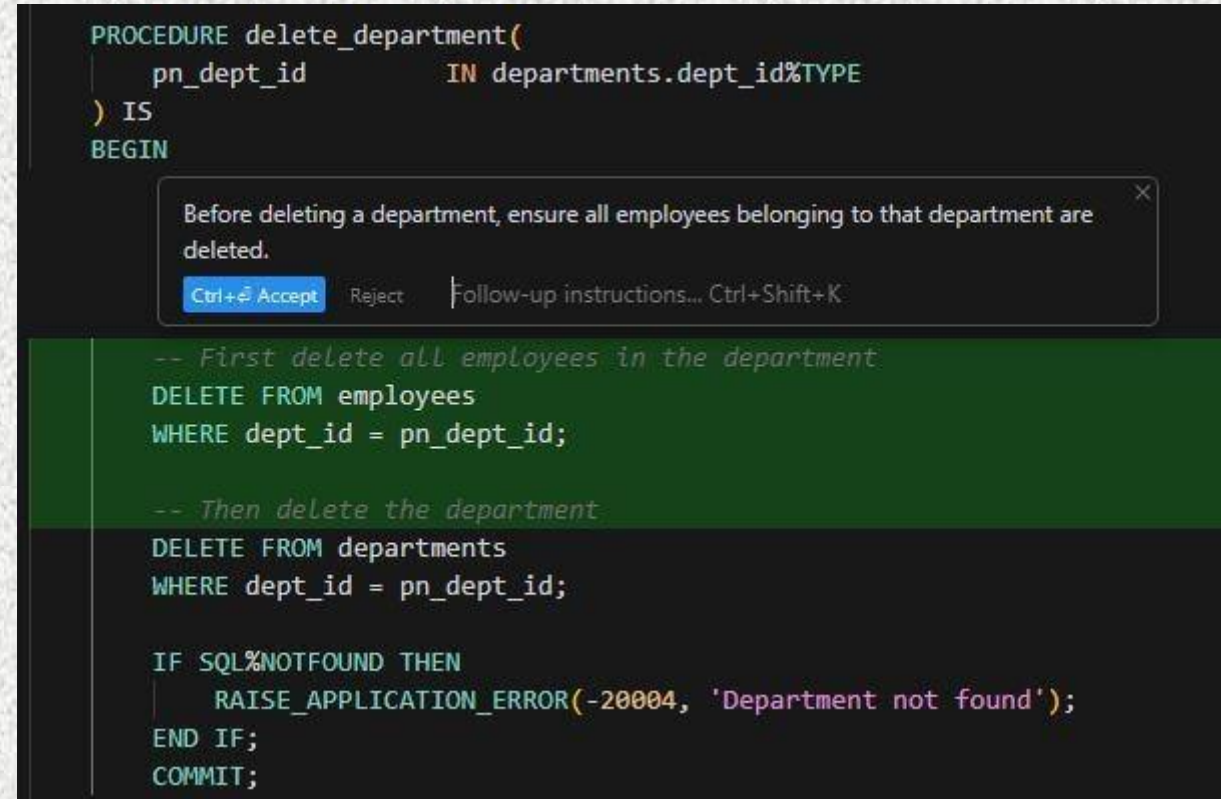


Control Cards for Agentic AI

Objective, Prior Work, & Limitations

Design Control Cards for agentic IDE to help users understand and steer LLM-proposed changes!

- TODAY: chat plus diffs, users accept/reject
- LIMITS:
 - Unclear intent, hidden scope, weak risk cues
- Mixed-initiative control matters (Horvitz, 1999)
- HAI guidelines stress correction & clarity (Amershi et al., 2019)
- Automation misuse occurs under over-trust (Parasuraman & Riley, 1997)



The screenshot shows a code editor with a dark theme. A SQL procedure named `delete_department` is visible. A control card overlay is present, displaying a warning message: "Before deleting a department, ensure all employees belonging to that department are deleted." The card includes buttons for "Accept" (with a keyboard shortcut `Ctrl+E`), "Reject", and "Follow-up instructions..." (with a keyboard shortcut `Ctrl+Shift+K`). The SQL code in the background includes comments and commands for deleting employees and the department, as well as an error handling block.

```
PROCEDURE delete_department(  
  pn_dept_id      IN departments.dept_id%TYPE  
) IS  
BEGIN  
  
  -- First delete all employees in the department  
  DELETE FROM employees  
  WHERE dept_id = pn_dept_id;  
  
  -- Then delete the department  
  DELETE FROM departments  
  WHERE dept_id = pn_dept_id;  
  
  IF SQL%NOTFOUND THEN  
    RAISE_APPLICATION_ERROR(-20004, 'Department not found');  
  END IF;  
  COMMIT;
```

What is new? Why will it work? Who cares?

➤ NEW:

➤ Card-first protocol:

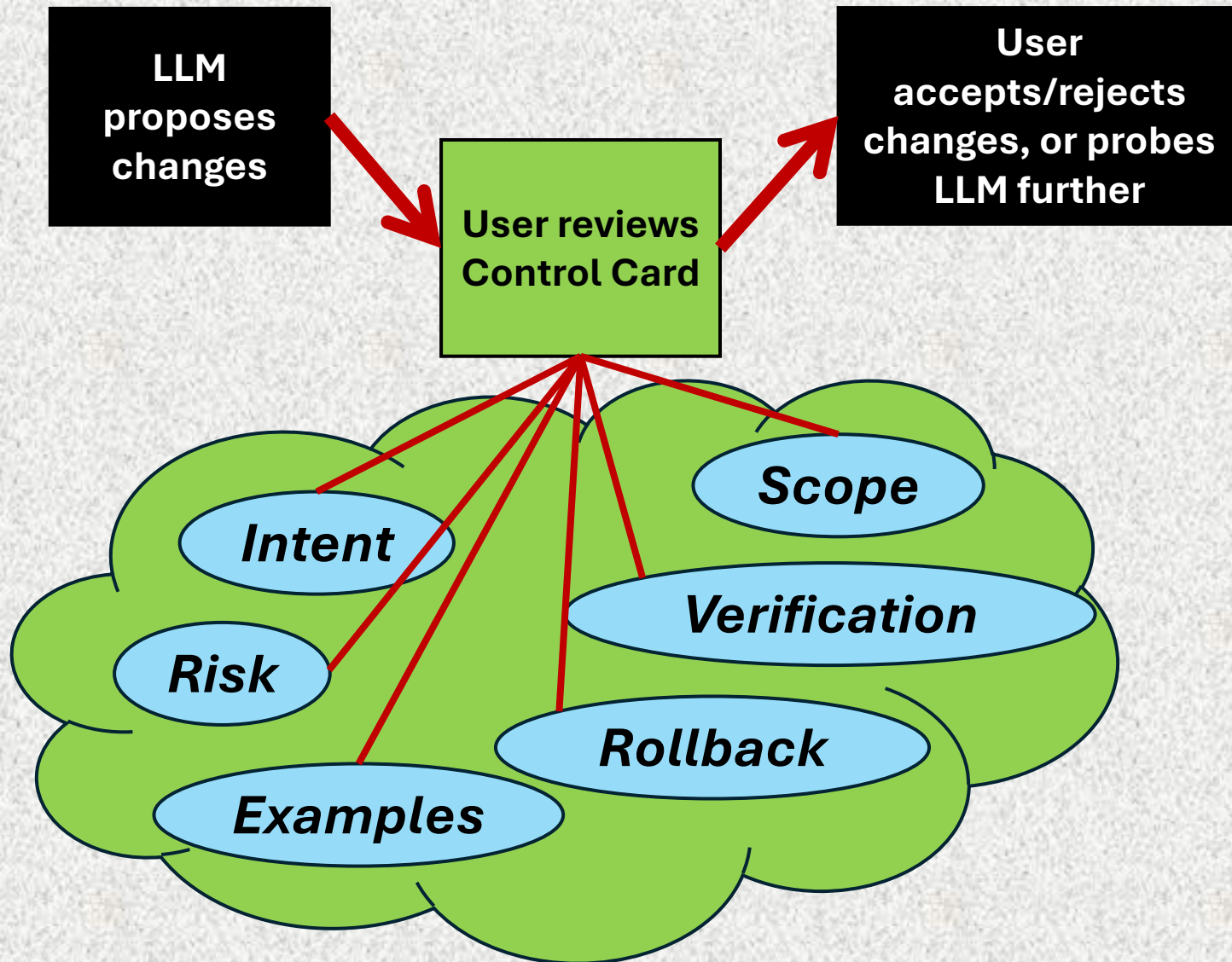
- No diff until user REVIEWS card and ACCEPTs changes

➤ Benefits:

- Increased control, transparency, & decision-making in Agentic IDEs
- Fewer over-trusted edits
- Modular, reusable HAI framework for safer AI workflows

➤ Users:

- Developers who may suffer from **automation bias**



Methods, Data, Expected Claims, Team

➤ Methods & Data:

- Implement using Cursor rules
- Run within-subject A/B for 10+ users (baseline vs Cursor rules)
- Use tasks with seeded traps and ambiguity
- Measure harmful edit accept rate, undos, clarifications
 - *Harmful edit: breaks tests, high security risk*

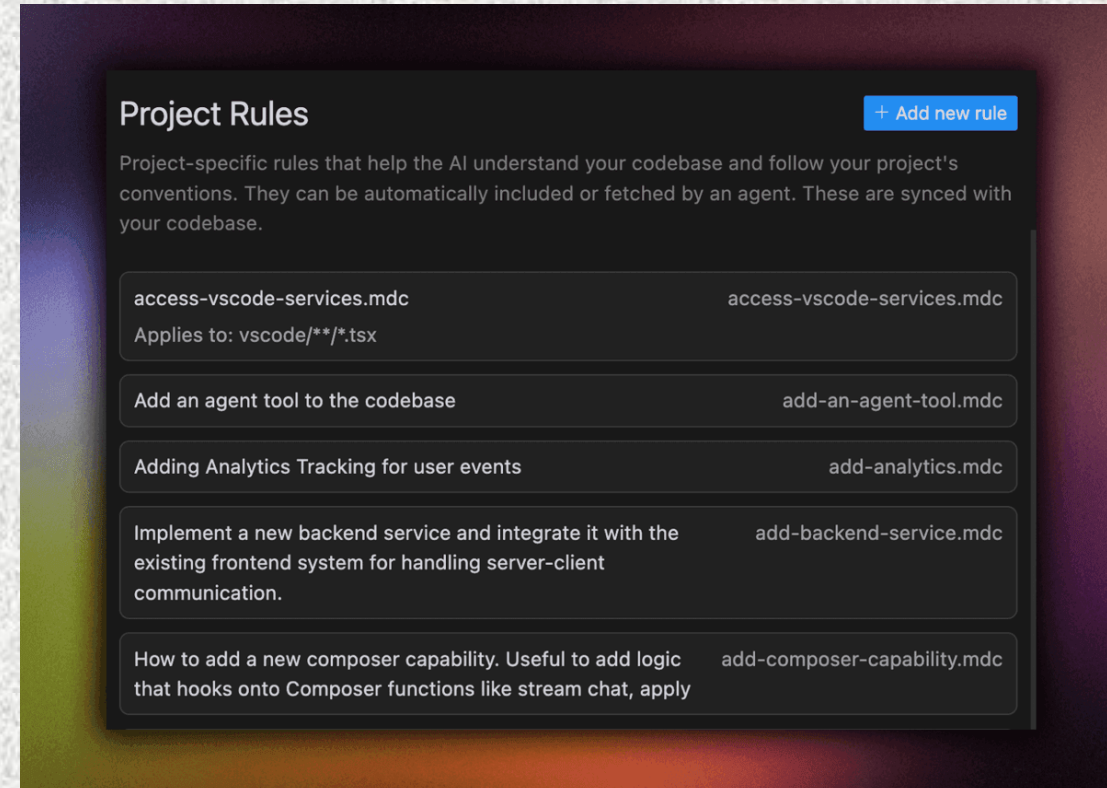
➤ Expected Claim:

- Demonstrate that control cards improve

HUMAN CONTROL

by reducing harmful AI edits efficiently!

➤ Team: Nirjhar Deb



References

- Amershi, S., Weld, D., Vorvoreanu, M., Fourney, A., Nushi, B., Collisson, P., Suh, J., Iqbal, S., Bennett, P. N., Inkpen, K., Teevan, J., Kikin-Gil, R., & Horvitz, E. (2019). Guidelines for human-AI interaction. In Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (Paper 3, pp. 1-13). ACM. <https://doi.org/10.1145/3290605.3300233>
- Horvitz, E. (1999). Principles of mixed-initiative user interfaces. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (pp. 159-166). ACM. <https://doi.org/10.1145/302979.303030>
- Parasuraman, R., & Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. Human Factors, 39(2), 230-253. <https://doi.org/10.1518/001872097778543886>